Topaz

Topaz builds software that learns in real time.

First product: a landing page that optimizes itself. Same substrate, next stop: edge and autonomous systems.

[PENDING-EMMA: confirm one-liner] · [PENDING-A1: A1 page-morphing loop as background]

THE PROBLEM

Most websites are too small to A/B test.

- One significant test needs ~400 conversions per variant: about 27,000 visitors at a 3% conversion rate.
- Below that line, the people who make funnels — indie SaaS founders, funnel and landing-page builders — guess, hand-tweak, or freeze whatever an AI page builder gave them.
- Today's "AI optimization" tools route traffic between human-written variants, and gate even that behind $199-$249/mo high-traffic tiers. Their statistics need the traffic the customer does not have.

WHY NOW

Generate-and-freeze, meet commodity embeddings.

- AI page builders commoditized page creation, and froze page improvement: millions of generated-and-frozen pages have no optimization path.
- The incumbent ceiling is structural (discrete variants), not a tuning gap that will close on its own.
- Frozen embedding spaces became free commodity infrastructure. That is exactly the substrate Sutra compiles onto.

THE SOLUTION

The page learns in real time, like an agent.

It improves itself as it runs, not from an outside agent guessing and re-testing it.

- The page IS a program in Sutra, compiled into a continuously adjustable tensor graph.
- Layout, spacing, emphasis, and color are continuous parameters.
- They learn from every scroll, hover, and dwell: dense behavioral signal, not binary conversions.
- So the page improves from the traffic it actually has — and stays readable: "what changed and why" is a real answer, not a heat map.

DEMO

A1: warmer / colder, the page morphing live.

Demo code complete and containerized (Docker; ~30-min deploy path). [PENDING-A1: live URL + recorded video — the two remaining artifacts]

EVIDENCE AND THE FOUR RISKS

We price down risk, then name the next pass/fail milestone.

Technical [retired] Sutra compiler ships (PyPI v0.9.4, five releases since May). NeurIPS submission: bundle decode 100% at k=8, bind/unbind error ~1.5e-15.

Product [mostly retired] A working OS prototype in the Topaz stack: terminal, calculator, GUI. The learner beats the best-variant bandit ceiling in simulation (0.824 at 500 visitors vs 0.609).

Commercial [open] Customer-discovery outreach wave live (Jul 2026), interviews against pre-written falsifiable hypotheses. Next signal: a structured pilot with named design partners.

Regulatory [deferred] None for the web wedge. It enters only with the edge / autonomous-systems markets deliberately sequenced behind it.

Next milestone (falsifiable): A2 pre-registered live experiment. Engagement-composite primary metric, ~250-500 post-consent visitors per arm, abort criteria fixed in advance, results published whatever they show. Pass = beats the frozen baseline on real traffic.

MARKET

A large, unserved beachhead, with a deep-tech road behind it.

- Beachhead ICP, kept deliberately narrow: the people who MAKE sales funnels — indie SaaS founders and funnel/landing-page builders. Self-serve, card-swipe sales, no procurement.
- Incumbents priced them out: continuous optimization appears only on $199-$249/mo high-traffic tiers; sub-500-visitor sites cannot buy it at any sane price. The segment is unserved, not underserved.
- Behind the wedge: edge and autonomous systems, where the whole system being one inspectable network becomes the product. Sequenced, not led with.

Serviceable beachhead $57M-$120M/yr: >=165k-200k paying customers of named funnel tools alone (ClickFunnels 150k disclosed 2023; Unbounce 15k+; Leadpages ~40k) x $29-$50/mo. Wider bottom-up TAM ~$7B/yr: ~24M US small-business sites (SBA 2024) x ~$300/yr.

Our own arithmetic from public anchors, assumptions stated in the plan; conservative bound uses only company-disclosed paying counts at the $29 floor.

WHY INCUMBENTS CAN'T FOLLOW

The ceiling is structural, not a tuning gap.

- Classical A/B tools need ~27K visitors per significant test. We learn at small-site traffic.
- AI variant routing (Optimizely-class) extracts ~1 bit per visitor and can never beat the best variant a human wrote.
- Topaz optimizes a continuous design space from dense per-visitor signal: a different sample-efficiency regime entirely.

WHAT'S UNDER IT

A language and an OS: the moat and the long game.

- Sutra: a geometrically compiled language; programs become differentiable tensor graphs. On PyPI (sutra-dev v0.9.4), NeurIPS 2026 submission.
- An OS prototype: a GPU-native neuro-symbolic operating system where the whole running system is one inspectable network.
- The page-as-program property, and with it the continuous optimization space, falls out of Sutra's compilation step. A competitor would have to rebuild the substrate to follow.

TRACTION AND FIRST COMMERCIAL SIGNAL

What exists, stated plainly.

- Shipped and public: Sutra compiler on PyPI (sutra-dev v0.9.4 — five releases May-Jul 2026), a working OS prototype, NeurIPS 2026 submission.
- In motion: customer-discovery outreach wave sent Jul 2026, interviews against pre-written falsifiable hypotheses; A1 demo code complete and containerized; A2 live experiment pre-registered.
- Pre-product, pre-revenue, no signed pilots yet. The strongest near-term signal we are building toward is a structured pilot with named design partners and written success criteria.

TEAM

Solo founder, full-time, operating with AI leverage.

- Emma Leonhart has shipped a working language compiler to PyPI, a working OS prototype (terminal, calculator, click-driven GUI), and a NeurIPS 2026 submission.
- The operating model is AI-leveraged under durable rules, queues, and automated work loops, with the judgment to know when to take the wheel.
- Third-party read, Y Combinator's Paxel analysis: "operating at a strong level... turning messy, multi-repo work into durable rules, releases, queues, cron loops, and truth-preserving engineering constraints."
- The case for backing one founder is this operating record. Hiring multiplies an already-leveraged operator; it does not replace a missing one.

THE ASK

Raising to prove the thesis on real traffic.

[PENDING-EMMA: round size + instrument. Tracks: YC + investors, with a cyber.fund-style uncapped SAFE in parallel.]

- Use of funds 1: run the A2 live-traffic experiment at scale (the falsifiable pass/fail milestone above).
- Use of funds 2: land the first structured pilots with named design partners and written success criteria.
- Use of funds 3: relocate operations to San Francisco (already decided).

Topaz: software that learns in real time.

Deep tech with a web wedge that proves the learning loop in public, on a falsifiable schedule — then the same substrate to edge and autonomous systems, behind a language-and-OS moat a competitor would have to rebuild to follow.

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